

WEEK 1 PRACTICAL

TASK 1

Develop a C program that demonstrates how a Linux operating system executes a command entered by a user that

1. Accept a Linux command as input.
2. Create a child process using `fork()`.
3. Execute the command in the child process using an appropriate `exec()` system call.
4. Allow the parent process to wait for the child using `wait()`.
5. Display the Process ID (PID) of both parent and child processes.

```
GNU nano 8.7.1
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>
#include <stdlib.h>

int main() {
    char command[100];
    pid_t pid;

    printf("Enter Linux command: ");
    scanf("%99s", command);

    pid = fork();

    if (pid < 0) {
        perror("fork");
        return 1;
    }

    if (pid == 0) {
        printf("Child PID: %d\n", getpid());
        printf("Parent PID: %d\n", getppid());

        execlp(command, command, (char *)NULL);

        perror("exec");
        exit(1);
    } else {
        printf("Parent PID: %d\n", getpid());
        printf("Child PID: %d\n", pid);

        wait(NULL);

        printf("Child process completed.\n");
    }

    return 0;
}
```

```
imvarls@LAPTOP-VQ4D70L7: ~$ nano week1.c
imvarls@LAPTOP-VQ4D70L7: ~$ gcc week1.c -o week1
imvarls@LAPTOP-VQ4D70L7: ~$ ./week1
```

```
imvarls@LAPTOP-VQ4D70L7: ~$ nano week1.c
imvarls@LAPTOP-VQ4D70L7: ~$ gcc week1.c -o week1
imvarls@LAPTOP-VQ4D70L7: ~$ ./week1
Enter Linux command: pwd
Parent PID: 566
Child PID: 567
Child PID: 567
Parent PID: 566
/home/imvarls
Child process completed.
imvarls@LAPTOP-VQ4D70L7: ~$ |
```

Task2

Using Linux terminal commands (uname, lscpu, lsblk, ps, top), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the OS abstracts CPU, memory, storage, and I/O devices.

1. Uname: Displays operating system and kernel information.

```
imvarls@LAPTOP-VQ4D70L7: ~$ uname
Linux
imvarls@LAPTOP-VQ4D70L7: ~$ |
```

lscpu: Displays CPU information.

```
Caches (sum of all):
  L1d: 288 KiB (6 instances)
  L1i: 192 KiB (6 instances)
  L2: 7.5 MiB (6 instances)
  L3: 12 MiB (1 instance)
NUMA:
  NUMA node(s): 1
  NUMA node0 CPU(s): 0-11
Vulnerabilities:
  Gather data sampling: Not affected
  Ghostwrite: Not affected
  Indirect target selection: Not affected
  Itlb multihit: Not affected
  L1tf: Not affected
  Mds: Not affected
  Meltdown: Not affected
  Mmio stale data: Not affected
  Old microcode: Not affected
  Reg file data sampling: Vulnerable: No microcode
  Retbleed: Mitigation; Enhanced IBRS
  Spec rstack overflow: Not affected
  Spec store bypass: Mitigation; Speculative Store Bypass disabled via prctl
  Spectre v1: Mitigation; usercopy/swapgs barriers and __user pointer sanitization
  Spectre v2: Mitigation; Enhanced / Automatic IBRS; IBPB conditional; PBRSE-eIBRS SW sequence; BHI BHI_D
  IS_S
  Smbds: Not affected
  Tsa: Not affected
  Tsx async abort: Not affected
  Vmscape: Not affected
```

lsblk: Displays storage devices.

```
imvarls@LAPTOP-VQ4D70L7:~$ lsblk
NAME MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda   8:0    0 356.9M  1 disk
sdb   8:16   0 159.4M  1 disk
sdc   8:32   0    2G    0 disk [SWAP]
sdd   8:48   0    1T    0 disk /snap
                                   /mnt/wslg/distro
                                   /

imvarls@LAPTOP-VQ4D70L7:~$ |
```

ps: Displays currently running processes.

```
imvarls@LAPTOP-VQ4D70L7:~$ ps
  PID TTY          TIME CMD
 1410 pts/3        00:00:00 bash
 1617 pts/3        00:00:00 ps
imvarls@LAPTOP-VQ4D70L7:~$ |
```

top: Displays real-time system performance.

```
top - 14:23:55 up 19 min, 1 user, load average: 0.00, 0.01, 0.00
Tasks: 29 total, 1 running, 28 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni,100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7782.6 total, 7062.3 free, 580.0 used, 292.4 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used, 7202.7 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1631	imvarls	20	0	8172	6040	3932	R	0.3	0.1	0:00.01	top
1	root	20	0	24572	15756	11568	S	0.0	0.2	0:01.21	systemd
2	root	20	0	3180	2204	2072	S	0.0	0.0	0:00.03	init-systemd(Ub
7	root	20	0	3180	2048	1940	S	0.0	0.0	0:00.00	init
47	root	19	-1	50392	16904	15600	S	0.0	0.2	0:00.29	systemd-journal
78	systemd+	20	0	22540	14548	11932	S	0.0	0.2	0:00.08	systemd-resolve
88	root	20	0	35364	12336	9244	S	0.0	0.2	0:00.67	systemd-udevd
108	root	20	0	2888	1948	1820	S	0.0	0.0	0:00.03	chronyd-starter
109	root	20	0	4472	3116	2868	S	0.0	0.0	0:00.02	cron
110	message+	20	0	8796	5448	4724	S	0.0	0.1	0:00.14	dbus-daemon
153	root	20	0	44092	29852	14712	S	0.0	0.4	0:00.22	networkd-dispat
158	root	20	0	18440	9440	8168	S	0.0	0.1	0:00.10	systemd-logind
177	syslog	20	0	220548	5964	4632	S	0.0	0.1	0:00.07	rsyslogd
188	root	20	0	5492	2736	2552	S	0.0	0.0	0:00.01	agetty
206	_chrony	20	0	24440	12008	7232	S	0.0	0.2	0:00.13	chronyd
207	root	20	0	123456	32540	17924	S	0.0	0.4	0:00.17	unattended-upgr
224	_chrony	20	0	12092	2436	1788	S	0.0	0.0	0:00.01	chronyd
325	root	20	0	8644	5496	4656	S	0.0	0.1	0:00.01	login
368	imvarls	20	0	22748	13500	11040	S	0.0	0.2	0:00.17	systemd
372	imvarls	20	0	22908	4088	2088	S	0.0	0.1	0:00.00	(sd-pam)
401	imvarls	20	0	6136	5400	3852	S	0.0	0.1	0:00.02	bash
759	root	20	0	306680	11620	1520	S	0.0	0.1	0:02.17	snapfuse
765	root	20	0	2072772	40672	25624	S	0.0	0.5	0:03.31	snapd
1014	root	20	0	233080	7420	1556	S	0.0	0.1	0:00.10	snapfuse
1176	root	20	0	159348	2228	1556	S	0.0	0.0	0:00.00	snapfuse
1227	imvarls	20	0	8544	5088	4736	S	0.0	0.1	0:00.01	dbus-daemon
1407	root	20	0	3184	1116	980	S	0.0	0.0	0:00.00	SessionLeader
1408	root	20	0	3200	1256	1108	S	0.0	0.0	0:00.01	Relay(1410)
1410	imvarls	20	0	6268	5544	3872	S	0.0	0.1	0:00.09	bash